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Statistical analysis of coastal bridge vulnerability based on empirical evidence from Hurricane Katrina

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This paper analyses bridge damage along the US Gulf Coast from the 2005 Hurricane Katrina. Most bridges were low-lying water crossings having multiple-span concrete superstructures with limited connectivity to the substructure. The paper emphasises statistical analysis of hazard-damage relationship for the Katrina exposed region. Multivariate logistic regression is conducted to identify which hazard intensities and bridge characteristics were strong predictors of damage level, considering such parameters as surge elevation, water speed, wind speed, relative surge, number of spans, and age of bridge. The results indicate that storm surge, as well as number of spans, correlates to increasing level of damage, and subsequent analyses evaluate this hazard intensity at the damaged and undamaged bridge locations. Empirical damage probability and damage state exceedance matrices are developed. Empirical fragility curves, estimating likelihood of being in or exceeding a damage state, are presented as a first step toward risk-based analysis of coastal bridges.

Keywords: bridge; coastal; storm surge; probabilistic analysis; fragility

1. Introduction

Coastal bridges are susceptible to a number of threats ranging from severe environmental conditions leading to corrosion and deterioration (Sun and Hong 2002), inundation during flooding (Malavasi and Guadagnini 2003) or scour (Muzzammil *et al.* 2008), susceptibility to liquefaction during earthquakes (Youd 1993) and storm-induced damage during hurricanes. The latter threat has been painstakingly realised in recent years – most notably during the 2005 Hurricane Katrina where bridges were revealed to be the most vulnerable critical component of the Gulf Coast transportation system. Over 44 bridges throughout Alabama, Louisiana and Mississippi, some with hundreds of affected spans, were damaged at an estimated cost of over US\$1 billion (Padgett *et al.* 2008). However, the reality of bridge damage during hurricanes is not isolated to the events along the Gulf Coast in 2005. Similar vulnerabilities have also been illustrated at Escambia Bay, Florida during Hurricane Ivan (2004) (Douglass *et al.* 2004), in Hokkaido, Japan during the Songda Typhoon (2004) (Okada *et al.* 2006), as well as in past decades during Hurricane Camille (1969).

Development in the US has continued to occur at a rapid pace along our nation's coast, with over half of the current population living within 80.5 km of the coast (Perry and Mackun 2001), and most of the nation's anticipated rapid population growth

occurring in emerging mega-regions in hazard prone locations (RPA 2005). Moreover, global climate change influencing sea level rise (ICF International 2007), intensity and frequency of storms could continue to yield a hazard to which thousands of existing coastal bridges may be increasingly susceptible (USDOT 2007). We continue to witness the regular destruction of our coastal bridge infrastructure, as revealed by the recent damage during Hurricane Ike (2008) to the Rollover Pass Bridge carrying state highway 87 near Galveston Bay, TX (Figure 1). These repeated events continue to punctuate the need to better understand the vulnerability of bridges susceptible to hurricane and storm surge events. Moreover, the wealth of recent empirical data, particularly the widespread and well-documented evidence provided by Hurricane Katrina, offers an opportunity to advance this understanding.

While recent research following Hurricane Katrina has targeted associated challenges, probabilistic bridge vulnerability assessment has not been addressed to date. Studies have focused primarily upon prediction of the hazard itself and modelling the storm surge (Chen *et al.* 2007, USDOT 2007), deterministic evaluation or testing of the surge and wave loading placed upon bridge decks (Douglass *et al.* 2006) and screening techniques for coastal bridges based on static capacity-demand checks (Sheppard and Dompe 2006).

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The use of empirical data to develop probabilistic bridge vulnerability estimates has been performed following other natural hazard events, such as earthquakes (Basoz *et al.* 1999, Yamazaki *et al.* 1999), yet never before enabled for hurricane-induced bridge damage due to a lack of data. Statistical analysis of the empirical damage data from Hurricane Katrina conducted in this paper offers new insight on the relationship between the hazard and likelihood of damage. These models are altogether lacking in nationwide packages, such as HAZUS-MH (FEMA 2005), and are the first step towards enabling regional risk assessment of coastal bridge and highway systems subjected to hurricane or storm surge events.

The following sections provide a summary of the bridge damage observed during Hurricane Katrina and data that will be used in the subsequent analysis, including estimates of the hazard at the bridge locations. Evaluation of the hazard–damage relationships is presented for different intensity measures (storm surge elevation, wind speed, etc.) both at the overall bridge system and deck component level.

2. Summary of bridge damage from Hurricane Katrina

Hurricane Katrina made landfall as a category 3 storm, roughly 64.4 km southeast of New Orleans, Louisiana, on 29 August 2005. Storm surges of 7.62–10.67 m were experienced along the coast in Louisiana and Mississippi, with maximum winds estimated at 53.6 meters/sec. Given the widespread catastrophic damage that Hurricane Katrina caused to the transportation infrastructure, a number of reconnaissance

investigations and surveys have been performed (NIST 2006, TCLEE 2006, Robertson *et al.* 2007, Okeil and Cai 2008). While several of these investigations highlighted both highway and railway bridges, this paper focuses on analysis of damage to highway-type bridges.

Common failure modes included span shifting or unseating, which are attributed to surge and wave loading. Associated component damage for the bridges with excessive hydrodynamic loading also included guardrail damage, bearing damage and spalling of bent beams, etc. An example of this type of failure is shown in Figure 2 for the Bay St. Louis Bridge in Mississippi, which required complete replacement. Impact from debris carried through the waterways, including barges or oil rigs, also resulted in span shifting and damage to piles. Other failure modes included damage to electrical and mechanical equipment, which is primarily attributed to inundation or wind damage to operator houses. Scour and undermining of foundation and approaches also occurred. In many cases, a combination of failure modes was reported, rendering the bridge non-functional for some period of time during repair or replacement (TCLEE 2006).

Many of the bridges damaged by Hurricane Katrina were composed, partially or entirely, of reinforced concrete decks without a design that would allow trapped air to escape in the event of high water. Moreover, the simply supported spans often had limited or no connectivity between the superstructure and substructure. Without sufficient resistance to lateral or uplift forces, many spans were shifted or unseated completely. The most notable instance of this



Figure 1. State highway 87 Rollover Pass bridge along the Texas coast with four shifted or unseated spans from Hurricane Ike (October 2008). Similar damage pattern continues to be exhibited as experienced in Louisiana and Mississippi in Hurricane Katrina (2005).



Figure 2. Shifted and unseated spans on the US-90 Bay St. Louis Bridge in Mississippi.

was the I-10 Bridge across Lake Pontchartrain (Figure 3a). This twin span structure had 473 spans shifted and 64 spans that were completely unseated and fell into the lake. The majority of the severe damage sustained by the bridges during Hurricane Katrina was to the superstructure. The hydrostatic and hydrodynamic loading imposed on these superstructures during hurricane-induced surge and wave action includes buoyancy, drag and inertial forces, forces associated with added mass and vertical slamming forces (Sheppard 2008). These translate into vertical uplift and horizontal loads that are a function of parameters such as surge elevation, wave height, relative deck elevation and deck geometry. Failure modes associated with surge-induced loading occurred in both traditional fixed-type bridges having continuous or simply supported spans, as well as in movable bridges having a swing, lift or bascule. The traditional non-movable spans in movable bridges suffered the typical modes of failure presented above, which are associated with storm surge-induced loads. In either bridge type, the damaged sections tended to be low-lying spans over water crossings.

Several improved design details or retrofits to protect coastal bridges from these failure modes are being considered based on the observed performance in Katrina. This includes increased elevation of spans, as was implemented in the replacement of the US-90 Bay St. Louis and Biloxi-Ocean Springs Bridges. Other considerations include providing air vents to reduce buoyancy forces, continuous spans to increase resisting loads, or providing resistive or sometimes sacrificial elements, such as shear keys or tie downs.

The relative performance of the I-10 Bridge across Lake Pontchartrain to the US-11 is a prime illustration of the differing performance of bridges at similar locations and subjected to similar hazards. Only minor damage was observed on the US-11 Bridge, including erosion at the abutment and drawbridge damage. The US-11 Bridge had continuity across the spans, and positive connectivity between the deck and bent beams was provided, as shown in Figure 3b, rather than simply supported spans resting on the bent beams. Vent holes, provided in the diaphragms, which help mitigate buoyant forces are also believed to be a contributing factor to the relatively small damage. Another example of contrasting performance is found between the US-90 Biloxi-Ocean Springs Bridge to an adjacent CSXT Biloxi Bay railroad bridge, as shown in Figure 4 and also noted in previous studies (Okeil and Cai 2008). The railroad bridge with 38.1 cm shear keys did not suffer span unseating, as did its highway bridge counterpart. Despite the array of strategies that may be used for future protection of bridges, the current study focuses on vulnerability of as-built structures without advanced coastal force accommodation systems. However, while material, construction type and characteristics such as year of construction or span length can be inferred from nationwide highway bridge inventory databases, future analysis and screening of vulnerable bridges would benefit from cataloguing of additional construction details, such as bearing type, provision for shear keys or vents, etc. These details could not be discerned for the thousands of bridges in the exposed region.



(a)



(b)

Figure 3. Contrasting performance of two nearby bridges: (a) I-10 twin spans versus (b) continuous spans of the US-11 over Lake Pontchartrain.

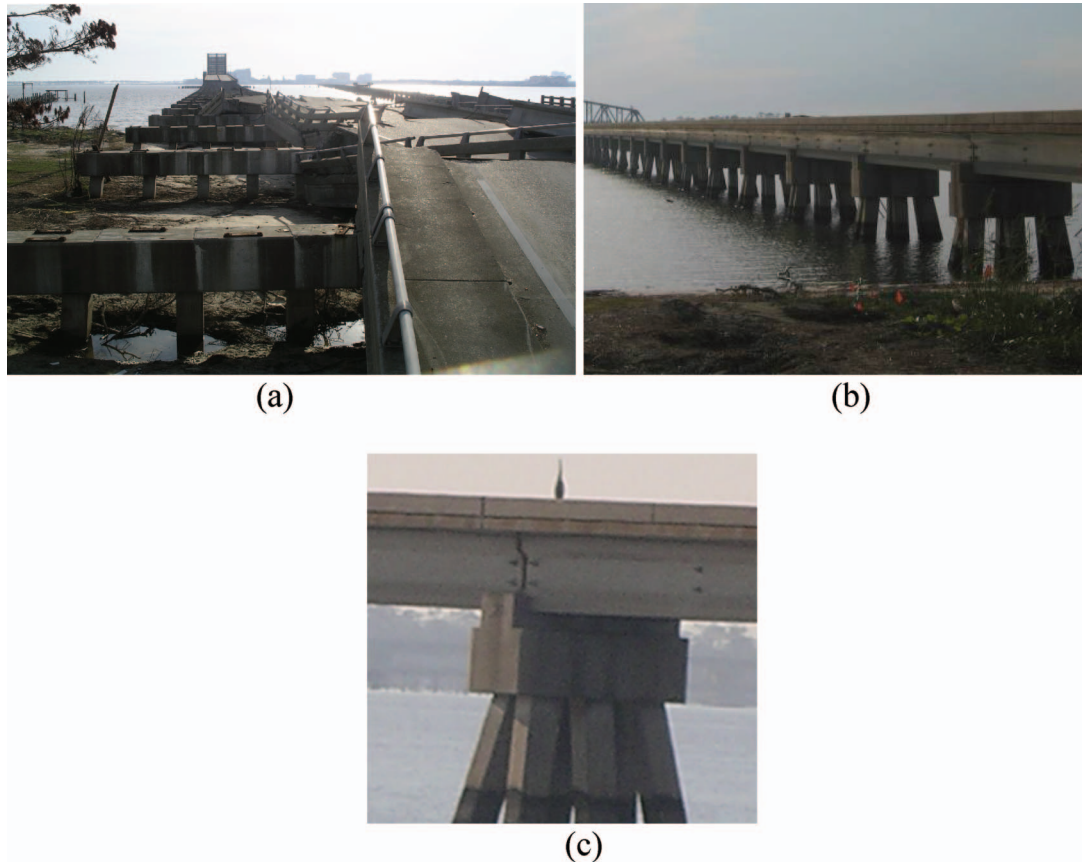


Figure 4. Contrasting performance of two nearby bridges: (a) US-90 Biloxi-Ocean Springs versus (b) adjacent railroad bridge, with (c) shear keys.

3. Data

Several datasets are used in this study to conduct statistical analysis of the bridge damage during Katrina. The first source of data is the empirical bridge damage data collected from the 2005 Hurricane Katrina reconnaissance supported by the American Society of Civil Engineers/Technical Council on Lifeline Earthquake Engineering (ASCE/TCLEE) and previously reported (TCLEE 2006). This dataset includes details of the 44 highway bridges damaged during the storm, including information such as the type and level of damage and repair cost estimates, as presented in Padgett *et al.* (2008). The same damage rating scale adopted includes the qualitative categories of slight, moderate, extensive and complete damage, as presented in Table 1. The HAZUS-MH (FEMA 2005) bridge damage scale typically used to describe seismic damage was amended to include additional failure modes associated with hurricane events, such as scour or submerged electrical/mechanical equipment. The number of highway bridges suffering each level of damage is presented in Figure 5a.

The national bridge inventory (NBI) database was consulted to locate bridge numbers, characteristics and

geographic coordinates for the damaged bridges, as well as for the total bridge set within the surge region (FHWA 2005). Surge estimates from Hurricane Katrina (FEMA 2006) are used in the subsequent analysis, which were developed and produced by FEMA using SLOSH modelling (NOAA 2007). Surge elevations affecting bridges in the surge zone were interpolated via geographic information system (GIS) from the surge contours. Figure 5b shows the FEMA surge elevation map in the Louisiana region, along with all of the bridges in the area, including damaged and undamaged bridges. An additional dataset employed during this study is the output of a Katrina hindcast developed by researchers at The University of Notre Dame (Westerlink *et al.* 2007). This dataset is used to enable additional intensity measures to be considered for the bridge locations, including estimates of water velocity and wind speed (10 minute average speed) resulting from an ADCIRC (Luetlich and Westerink 2004) model. The model includes a nodal mesh of the area in the path of the hurricane and was made up of over 2,000,000 nodes. The nearest node to each of the 2122 bridges in the affected region is identified to estimate the storm intensity at the bridge locations. The boundaries of

Table 1. Bridge damage states adapted from Padgett *et al.* (2008).

Damage rank	Damage state	Description of associated damage
1	Slight	Minor cracking and spalling to the abutment, cracks in shear keys at abutments, minor spalling and cracks at hinges, minor spalling at the column (damage requires no more than cosmetic repair), minor cracking to the deck, or slight damage to operator house.
2	Moderate	Any column experiencing moderate (shear cracks) cracking and spalling (column structurally still sound), moderate movement of the abutment (<5.1 cm), extensive cracking and spalling of shear keys, any connection having cracked shear keys or bent bolts, keeper bar failure without unseating, rocker bearing failure, moderate settlement of the approach, moderate scour of the abutment or approach, damage to guardrails, wind and/or water damage to operator house resulting in switchboard or content damage.
3	Extensive	Any column degrading without collapse – shear failure (column structurally unsafe), significant residual movement at connections, or major settlement approach, vertical offset of the abutment, differential settlement at connections, shear key failure at abutments, extensive scour of abutments, or submerged electrical or mechanical equipment.
4	Complete	Any column collapsing or connection losing all bearing support, which may lead to imminent deck collapse, span unseating, or tilting of substructure due to foundation failure.

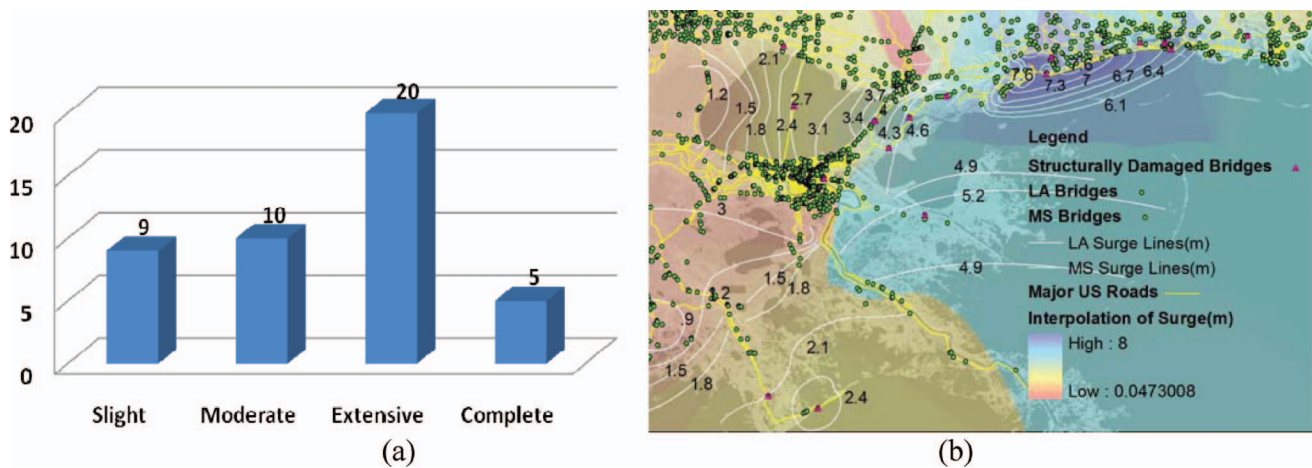


Figure 5. (a) Number of highway bridges in each damage state. (b) Example of bridge damage data (damaged bridges shown in pink; undamaged bridges shown in green) relative to surge elevation contours in Louisiana.

analysis for the exposed bridges are taken as the Gulf of Mexico on the southern side, 30.8° N on the northern side and then 87.8° W and 90.66° W on the eastern and western sides respectively. Due to the fact that only peak navigation clearance is documented in the NBI for major water crossings, yet not for approach span elevation, relative surge (surge bridge elevation) is not feasible or reliable as a hazard intensity measure for the entire exposed inventory of hundreds of bridges. Relative surge is considered only for the damaged bridges using the predominant deck elevation from reconnaissance reports or inferred from the approach roadway elevation per Department of Transportation (DOT) recommendation (according to S. Sibley, Louisiana Department of Transportation (LADOT), personal communication, 2008). For many of the long-span water-crossing bridges in the region, the road or embankment elevation is maintained until the main

spans are reached and is estimated for the study bridges from the USGS elevation maps (USGS 2008).

Upon studying the damaged bridges, several obvious patterns are noticed. The first significant similarity was the service type of the bridges involved, in addition to the construction material of the damaged spans previously noted. All but one of the damaged bridges were water crossings. The one bridge that did not cross water crossed a railroad and was immediately adjacent to water. Therefore, in the statistical analysis, emphasis is placed upon the bridges over water with concrete superstructures, rather than the total number in the overall region. Moreover, this bridge population is limited to low-lying bridges due to the dominance of this characteristic for the damaged bridges; thus, bridges over navigable waterways with inferred approach elevations greater than 9.14 m are not included. A total of 262 bridges are considered in the analysis. This results

in a more homogeneous dataset and a reduced total number of bridges to be analysed in the exposed population. Statistical analysis of the damaged and undamaged bridges is presented in the following section.

4. Hazard–damage analysis

The subsequent analysis aims to assess the empirical probability density and cumulative distribution functions for the hazard intensity at the damaged bridge locations, as well as identify additional bridge characteristics and hazard intensities that contribute most to increasing damage. The level of bridge damage relative to the hazard intensity is evaluated in order to assess potential damage–hazard correlation. The second phase of analysis considers the entire exposed population of bridges in order to provide damage probability estimates for coastal bridges based on the empirical data.

4.1. Damage distribution

The damaged bridges are ranked 1 through to 4, corresponding to slight through to complete damage, and compared to each of the hazard intensities evaluated in this study. Forty of the damaged bridges had sufficient hazard–damage data for use in the analysis. The four intensity measures include peak storm surge elevation, relative surge elevation (surge–deck elevation), water speed and 10 minute averaged wind speed. These intensity measures are assessed due to their availability from hindcast or inferential data for Katrina and their potential relation to structural

damage. For example, as previously noted, the surge elevation influences the hydrodynamic and hydrostatic loading on bridge decks; the water speeds affect the hydraulic loading on submerged structural elements and the potential speed at which debris may impact the bridge; and the wind speeds not only affect the surge and wind loading, but also the wave height. Figure 6 shows the damage rank relative to the estimated hazard intensities for the damaged bridges. This form of analysis emphasises the continuous nature of the intensity measures, despite categorical damage rank, and gives an indication of the correlation between level of damage and intensity measure. As illustrated in the figure, there is little correlation between increasing damage level and water speed or 10 minute average wind speed. However, a stronger and perhaps more intuitive relationship is exhibited between damage rank and peak surge elevation. Therefore, the results of the later analyses will emphasise this intensity measure.

The hazard intensity measures and characteristics of the damaged bridges are analysed further through multivariate logistic regression. This analysis is conducted to identify the characteristics that contribute most to the observed level of bridge damage, and is commonly performed for discrete outcome situations. In this case, the discrete dependent variable is the damage state classification, deemed slight, moderate, extensive or complete. The model effects, or independent variables, considered in the analysis include peak surge elevation, relative surge, water speed, averaged wind speed, number of spans and year built. These

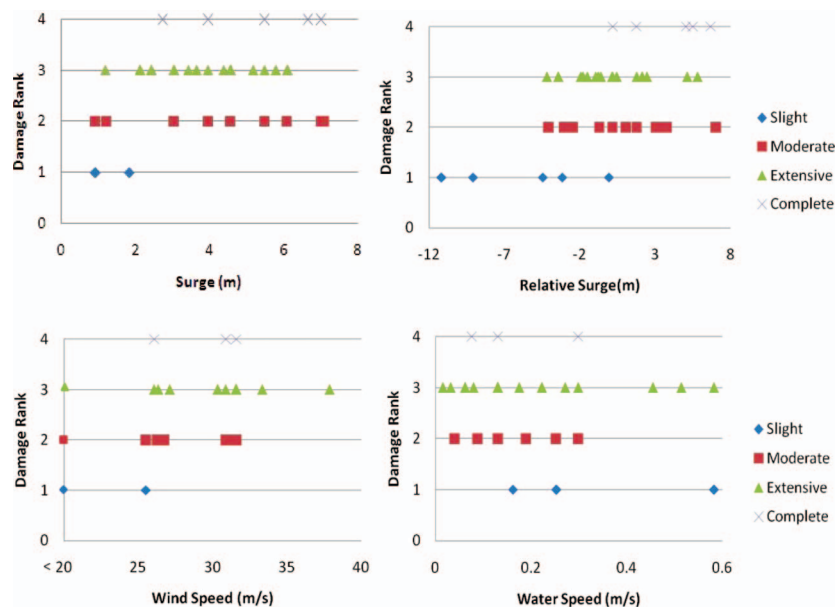


Figure 6. Damage rank relative to the hazard intensity measure at the bridge location. Note that the wind speed is the 10 minute average speed, and the relative surge is the estimated surge deck elevation.

additional characteristics of the damaged bridges are shown in Figure 7, including a categorised breakdown of the number of spans and year built, although these are treated as continuous variables in the correlation analysis. The software package JMP is used to conduct the logistic regression. The analysis reveals that the most important parameters are the surge elevation, number of spans and relative surge, with p -values of <0.001 , <0.001 and 0.019 , respectively. The findings are consistent with expectations from visual inspection of the hazard–damage relationship in Figure 6, which indicate a potential correlation between damage level and surge elevation or relative surge. Identification of the number of spans as an important parameter based on the logistic regression may be attributed to the fact that the longer bridges with more spans spanning open waterways may suffer a greater exposure to the hazard and have a potential for more damage in at least part of the bridge. This also indicates that further binning of the bridges by number of span may be prudent in the future should sufficient empirical or simulated data become available. The year built is a characteristic of interest in part because of its relation to evolving design codes and construction practices. In addition, the year built can be related to degradation or duration of environmental exposure, such as that which leads to

corrosion of reinforcement steel or bearings, thereby degrading the structural resistance to extreme loading. However, the assessment reveals that the year built is not strongly correlated to damage level.

Histograms are developed to present the number of bridges meeting or exceeding each damage state. The damaged bridges are binned by intensity measure of the hazard, such as level of peak storm surge elevation due to its identification as a damage predictor. Figure 8 shows the histogram for bridges that suffered at least slight, moderate, extensive and complete damage, binned by storm surge elevation. The empirical cumulative distribution function is plotted on the same figure. This analysis indicates that 50% of the bridges suffering moderate damage or greater were subjected to a storm surge of at least 4 m. Similarly, when considering all of the damaged bridges (those suffering at least slight damage), the median value of the surge is approximately 3 m, indicating half of the damaged bridges were subjected to at least 3 m of surge. This complements the information gleaned from the previous plots in Figure 6, which keep the bridges distinguished by damage rank. They indicate, for example, that all of the bridges specifically classified as slightly damaged had surge elevations of 1.83 m or less.

The component damage investigation is focused on bridge decks—specifically number of spans damaged. Several of the damaged bridges in Katrina had hundreds of shifted or unseated spans. Table 2 lists the total number of shifted or unseated spans for each bridge that suffered span displacement, relative to other bridge characteristics previously discussed. For example, the I-10 Bridge across Lake Pontchartrain had 473 spans shifted and 64 spans that were completely unseated and fell into the lake. In total, analysis of the damaged highway bridges reveals that approximately 976 bridge spans were either shifted off their supports or unseated and cast into the water below during Hurricane Katrina. This count does not include damage to the movable spans. A summary of the shifted and unseated spans relative to peak storm surge elevation is shown in Figure 9. This indicates that damage to the spans in terms of shifting or unseating did not tend to occur until the surge reached elevations of approximately 3 m. Nearly 80% of the shifted or unseated spans occurred at surge elevations of 5 m or less.

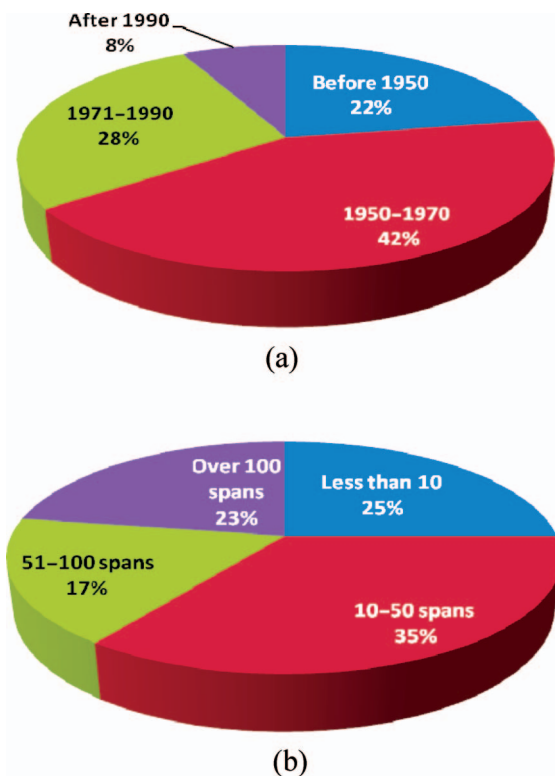


Figure 7. Distribution of (a) year built and (b) number of spans for the damaged bridges considered in the analysis.

4.2. Empirical fragility estimates

The empirical data presented above is used to develop estimates of the conditional probability of meeting or exceeding different damage states, given surge elevation. Point estimates of the exceedance probability can be derived by directly comparing the hazard at the

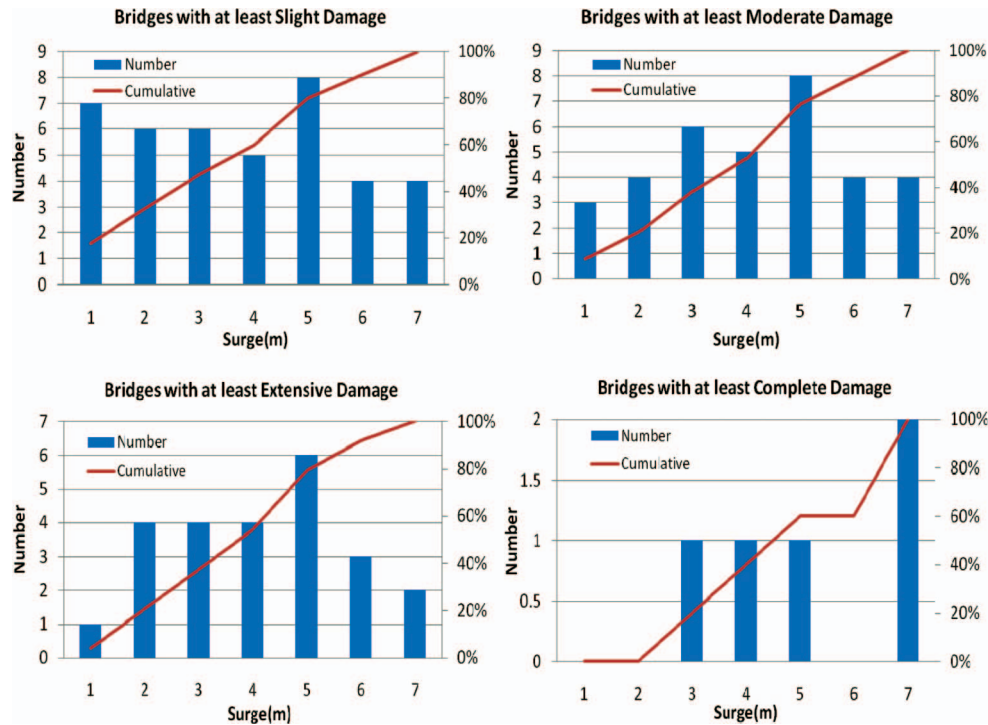


Figure 8. Histograms and empirical cumulative distribution functions of the peak surge elevation of bridges meeting or exceeding a damage state.

Table 2. Characteristics and number of spans displaced for bridges having considerable residual shifting or unseating of spans.

Bridge name	Facility	State	Bridge type	Damage state	Surge (m)	Year built	Spans [†]	Spans shifted	Spans unseated
Lake Pontchartrain	I-10	LA	MSSS-PC	Complete	3.96	1963	872	473	64
US-90 Bay St. Louis	US-90	MS	MSSS-RC + M	Complete	5.49	1953	243	1	242
Biloxi-Ocean Springs	US-90	MS	MSSS-RC + M	Complete	6.66	1961	153	2	129
Pontchartrain Causeway	LA Causeway	LA	MSSS-PC	Complete	2.74	1956	3002	0	17
Caminada Bay	LA-1	LA	MSSS-RC	Extensive	2.44	1961	110	13	0
Popps Ferry	Popps Ferry Rd.	MS	MSSS-RC + M	Extensive	5.79	1976	69	13	0
Henderson Point	US-90	MS	MSSS-PC	Complete	7.01	2000	36	6	1
I-10 Pascagoula River	I-10	MS	MSC-PC	Extensive	4.63	1976	660	6	1
Mobile Delta Causeway	I-10 / US-90	AL	MSSS-PC	Extensive	3.05*	1978	28	5	0
David V. LaRosa	W. Wittman Rd.	MS	MSSS-PC	Moderate	7.01	1970	20	3	0

Note: MSSS = multi-span simply supported; MSC = multi-span continuous (~6 span continuity); RC = reinforced concrete; PC = prestressed concrete; + M = also includes a movable span; [†]number of spans includes total spans if a twin structure; *surge estimate if outside FEMA contour zone.

damaged bridge locations relative to undamaged bridges, and are presented in the form of damage probability matrices. As noted previously, a homogeneous bridge sample is preferred, and the total number of bridges considered in the limited dataset is 262, including 36 of the damaged bridges, considering only those in Mississippi and Louisiana, where sufficient hazard data is available at both the damaged and undamaged bridge locations.

Table 3 presents the damage matrix of the bridges analysed in the Katrina exposed region along the Gulf Coast. This table shows the number of bridges within

each damage state at various storm surge elevations for the low elevation concrete water-crossing bridges. The complementary data in Table 4 shows the empirical damage probability matrix, which is the probability of being in a damage state for a particular surge elevation. This is a direct representation of the damaged and undamaged bridges in each bin, and can be derived directly from the data presented in Table 3.

The damage state exceedance probability matrix is presented in Table 5. The matrix offers percentiles of the cumulative distribution function for the bridge

fragility (probability of being in or exceeding a damage state conditioned upon hurricane-induced storm surge elevation). The potential uncertainty in each estimate and bias due to limited data for each bin is acknowledged. Collection of data from additional storms, as well as refinement of bridge inventory databases to include additional parameters, can help to overcome these challenges. The parameters of the fragility curve for each damage state are estimated following a lognormal distribution using a simple linear regression. The resulting empirical fragility curves for hurricane and storm surge damage to bridges from Katrina is shown in Figure 10. The slight damage state was not found to be a good fit and is not included in the figure.

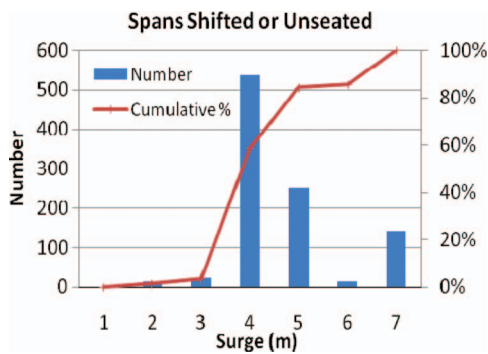


Figure 9. Histograms and empirical cumulative distribution functions of the peak surge elevation of component damage for spans.

The fragility curves provide an estimate of the probability of meeting or exceeding a damage state (moderate, extensive or complete), given a storm surge estimate, for a class of low elevation, multi-span simply supported concrete water-crossing bridges. The resulting empirical fragility curves indicate, for example, a median value of surge elevation of approximately 7.7 m for the moderate damage state, and also illustrate a relatively high dispersion for the class of bridges examined. The fragility estimates shown in Figure 10 can be used in validation of future analytical studies that simulate bridge damage from surge and wave action in hurricanes to provide refined fragility estimates. As previously noted, uncertainty in the fragility estimates presented herein is introduced by the limited damage data available from Katrina regional bridge inventory data, and is biased by the storm characteristics. Future work using analytical models and simulated data could offer the opportunity for enhanced binning and uncertainty treatment of bridge classes, by discerning details such as connection type and binning by number of spans, and refining relative surge elevation estimates for damaged and undamaged bridges. Additionally, enhanced uncertainty treatment at the hazard characteristic level would be afforded by simulated samples across a range of wave height/period parameters or using joint hazard intensity measures. Such probabilistic models of bridge failure form the foundation for risk assessment of coastal bridge infrastructure, and an opportunity to enhance regional risk mitigation along the coast.

Table 3. Damage matrix for bridges considered in Mississippi and Louisiana Katrina exposed region.

Observed damage state	Surge elevation (m)						Total
	0.61–1.83	1.83–3.05	3.05–4.27	4.27–5.49	5.49–6.71	6.71–7.92	
None	71	59	35	23	27	11	226
Slight	4	2	0	0	0	0	6
Moderate	2	0	3	1	0	2	8
Extensive	1	4	4	5	3	0	17
Complete	0	1	1	0	2	1	5
Total	78	66	43	29	32	14	262

Table 4. Damage probability matrix (probability of being in a damage state).

Observed damage state	Surge elevation (m)					
	0.61–1.83	1.83–3.05	3.05–4.27	4.27–5.49	5.49–6.71	6.71–7.92
Slight	0.051	0.030	–	–	–	–
Moderate	0.026	–	0.070	0.034	–	0.143
Extensive	0.013	0.061	0.093	0.172	0.094	–
Complete	–	0.015	0.023	–	0.063	0.071

Table 5. Exceedance probability matrix depicting empirical probability of being in or exceeding a damage state for a given surge elevation.

Damage state exceeded	Surge elevation (m)					
	0.61–1.83	1.83–3.05	3.05–4.27	4.27–5.49	5.49–6.71	6.71–7.92
Slight	0.090	0.106	0.186	0.207	0.156	0.214
Moderate	0.038	0.076	0.186	0.207	0.156	0.214
Extensive	0.013	0.076	0.116	0.172	0.156	0.071
Complete	–	0.015	0.023	–	0.063	0.071

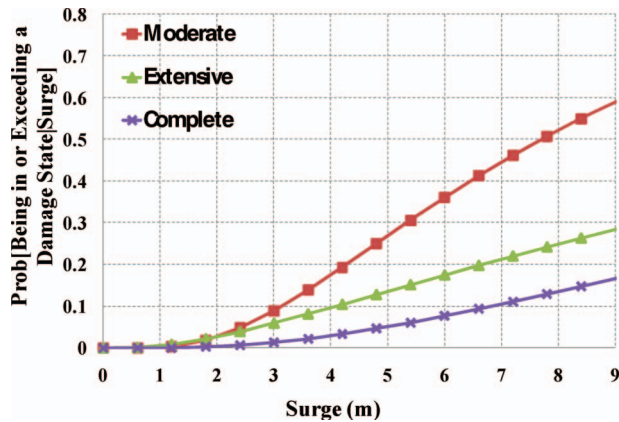


Figure 10. Fragility curves for bridge damage (probability of being in or exceeding a damage state) conditioned upon surge elevation, based on empirical data from Hurricane Katrina.

5. Conclusions and opportunities

Hurricane Katrina revealed the potential for significant damage to coastal bridges subjected to hurricane and storm surge events, with 44 damaged highway bridges in Mississippi, Alabama and Louisiana. A majority of the severe structural damage was attributed to hydraulic loading leading to span shifting and unseating, guardrail and bearing damage. Additional bridge damage was attributed to inundation of electrical and mechanical equipment in movable spans and impact from debris, among others documented in recent reconnaissance reports. Not including damage to movable spans, it is estimated that there were a total of 522 spans shifted and 454 spans unseated during Hurricane Katrina. Assessment of the damaged bridges reveals that most of the bridges were multiple span bridges with simply supported concrete superstructures. These bridges tended to be low-lying bridges over waterways. Of the bridges analysed, construction years ranged from 1930 to 2004, although nearly 50% of the bridges were constructed prior to 1970. This is, however, fairly representative of the overall inventory and was not found to be a strong predictor of damage level.

A multivariate logistic regression was conducted to evaluate which hazard intensities and bridge characteristics were important parameters in predicting increasing level of bridge damage (slight, moderate, extensive or complete). While the relationship between surge elevation, relative surge, water speed and mean wind speed, were all considered, only the surge related hazard parameters showed a strong correlation to damage level. In addition, the number of bridge spans was found to be closely related to damage level, which may be attributed to longer bridges over open waterways having a higher exposure and damage potential. This also indicates that binning by number of spans may be preferable in the future should sufficient empirical or simulated data exist. The inclusion of additional parameters in nationwide bridge databases could improve future analyses, such as additional fields for bearing type, coastal construction details (i.e. shear keys, air vents, etc.) and deck elevation over mean water height, which had to be inferred in the present study. Given the correlation between surge elevation and bridge damage level, empirical frequency and cumulative distribution functions were developed for the damaged bridges. This analysis revealed a median value of bridges suffering at least slight damage of 3 m, indicating that over half of the damaged bridges suffered peak surges over 3 m, while half of the bridges suffering at least extensive damage faced surges estimated at over 4 m. The bridge component level analysis indicated that span damage tended to occur at surge elevations of 3 m or greater, while nearly 80% of the span damage occurred by 5 m.

Probabilistic analysis of the damage versus undamaged bridges in the region offers empirical damage probability matrices (probability of being in a damage state) and exceedance probability matrices (probability of being in or exceeding a damage state). Regression analysis using this information provides the first set of fragility curves for bridges subjected to hurricane-induced storm surge hazards. These statements of conditional probability of failure can be used in estimating damage potential for similar coastal bridges, and for validating future analytical fragility estimates. Future work using analytical models should

address further sub-classification of bridges with increased simulated data, simulation of variation in hazard parameters height to avoid event bias and assessment of multiple intensity measures such as surge and wave height. Given the crucial importance of a functioning transportation system following natural hazard events, research of this nature forms a critical foundation for regional risk assessment and loss estimation in order to inform mitigation planning and promote transportation system sustainability.

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